GEOPHYSICS, VOL. 72, NO. 2 (MARCH-APRIL 2007); P. V51–V58, 11 FIGS.
10.1190/1.2431635

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Stacking of surface waves

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ABSTRACT

The seismic surface wave method (SWM) is a powerful means of characterizing near-surface structures. Although the SWM consists of only three steps (data acquisition, determination of dispersion curves, and inversion), it is important to take considerable care with the second step, determination of the dispersion curves. This step is usually completed by spectral analysis of surface waves (SASW) or multichannel analysis of surface waves (MASW). However, neither method is ideal, as each has its advantages and disadvantages. SASW provides higher horizontal resolution, but it is very sensitive to coherent noise and individual geophone coupling. MASW is a robust method able to separate different wave types, but its horizontal resolution is lower. Stacking of surface waves (SSW) is a good compromise between SASW and MASW. Using a reduced number of traces increases the horizontal resolution of MASW, and utilizing other shot records with the same receivers compensates for the decreased signal-to-noise ratio. The stacking is realized by summing the f-k amplitude spectra of windowed shot records, where windowing produces higher horizontal resolution and stacking produces improved data quality. Mixing is applied between the stacks derived with different parameters, as different frequency ranges require different windowing. SSW was tested and corroborated on a deep seismic data set. Horizontal resolution is validated by x-v plots at different frequencies, and v-f plots present data quality.

INTRODUCTION

Traditional seismic techniques are based on body-wave propagation, and in this respect, surface waves (the highest energy wave types) are coherent noise that must be eliminated. However, when processing surface waves, we reverse this view and consider the ground roll signal and any other wave type (e.g., reflections, refractions, and diffractions) to be coherent noise.

The high-frequency components of surface waves are confined to the near-surface, lower-velocity materials, although the low frequencies penetrate the higher-velocity materials at depth. Thus, the dispersion curve shows phase-velocity decreasing as frequency increases.

Surface waves have been utilized for seismological applications since the end of the 1950s, and Rayleigh waves have been widely used since the 1980s, when the spectral analysis of surface waves (SASW) was introduced (Nazarian and Stokoe, 1984). A single pair of receivers is configured and reconfigured with different offsets as many times as necessary to sample the desired wavelength (depth) range. Data are analyzed in the frequency domain to produce a dispersion curve by calculating the phase difference between each receiver pair. This method has been used for a wide variety of geotechnical and geoengineering applications (Stokoe et al., 1994). Although SASW has a high horizontal resolution, it is very sensitive to other wave types. The method is illustrated for a single pair of receivers by the flowchart in Figure 1a. The recorded pair of traces is transformed into the frequency domain by fast Fourier transform. Cross power spectrum (G_{xy}), phase difference (Φ), time delay (t), and finally, Rayleigh velocity (V_R), is calculated at each desired frequency (wavelength).

In spite of its lower horizontal resolution, multichannel analysis of surface waves (MASW) overcomes the weakness of the SASW method because it enables a robust and stable evaluation of subsoil properties. Many authors have determined the dispersion curves by MASW: Nolet and Panza (1976) calculated dispersion curves by f-k wavefield transformation; McMechan and Yedlin (1981) applied the τ-p transform; and Park et al. (1999) used a swept-frequency record by convolving an impulsive record with a stretch function. The simplified processing flow of MASW is shown in Figure 1b. The shot record (with off-end configuration, where the source is outside the range of receivers) is transformed into the f-k domain, and amplitude spectra are calculated. This data set is transformed into the v-f domain, $F_2(v,f) = F_1(f,f/v)$, where v is the velocity, and relative and absolute maxima are chosen at each frequency. Finally, points are selected to represent the $V_R(f)$ for different modes whenever the modes are separable.

Manuscript received by the Editor January 25, 2006; revised manuscript received September 11, 2006; published online February 28, 2007.
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Stacking of surface waves (SSW) is based on the MASW method. Deep seismic records presented in the paper were used to test the method. This measurement was made by low-frequency single geophones using a high-energy explosive source. The resulting large amount of data (extensive records in space and time) enables one to separate the different wave types and several modes of surface waves. SSW optimizes the windowing of input records. Windowing the data produce a better horizontal resolution, and the stacking of *f-k* spectra improves data quality.

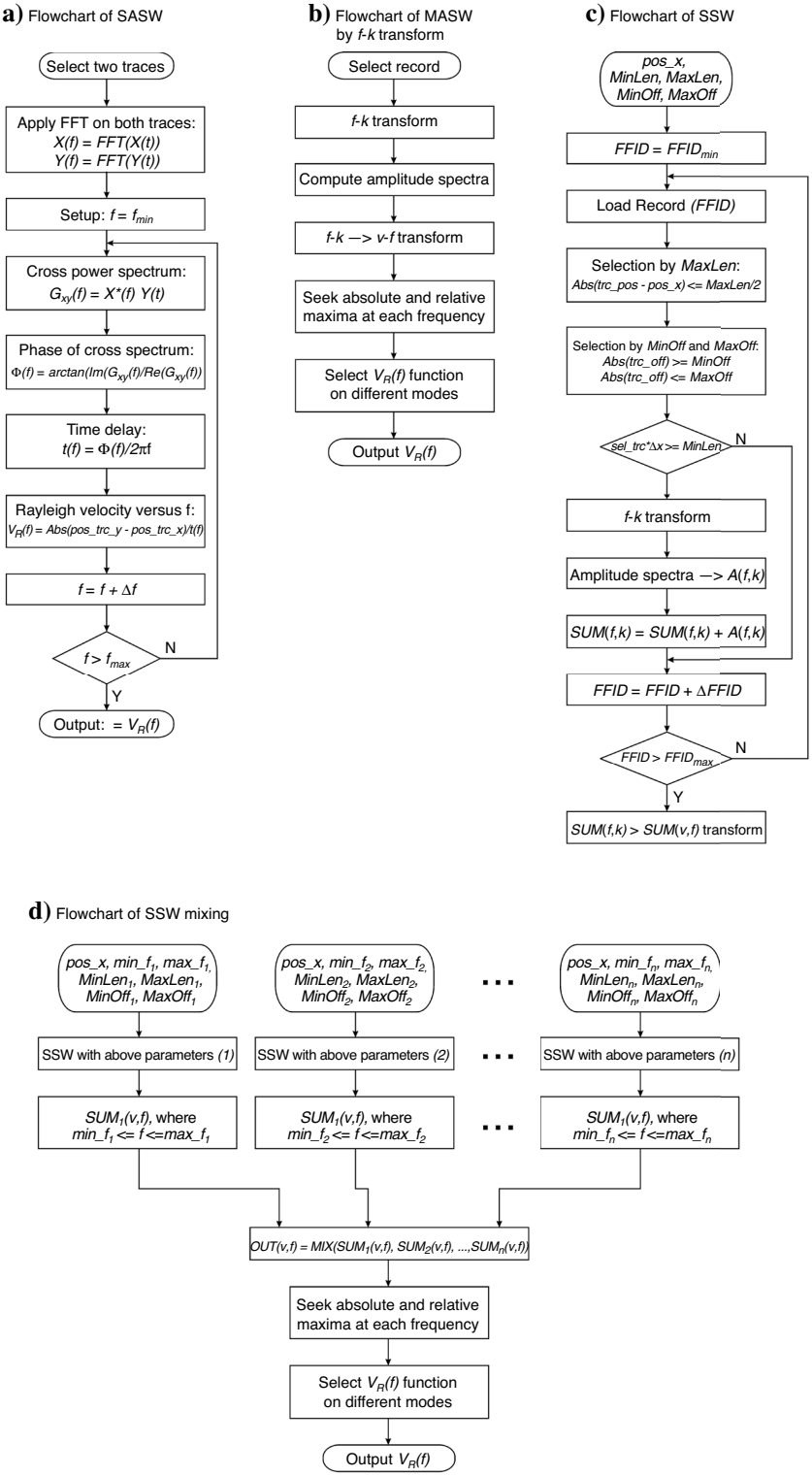

Figure 1. (a) Flowchart of the SASW method; (b) simplified process of the MASW method; (c) flowchart of the SSW method; and (d) flowchart of the mixing of *v-f* plots. (X^* denotes the complex conjugate of X).

THE PROPOSED SSW METHOD

SSW tries to overcome the only weakness of MASW (lower horizontal resolution compared with SASW) by windowing records in the space domain, and it makes an attempt to increase the data quality by stacking. For several shots, traces from a window around an individual receiver station are converted into the *f-k* domain, and the calculated amplitude spectra are stacked. A better *f-k* amplitude spectrum can be obtained by stacking the appropriate *f-k* panels. Finally, the *v-f* plots are derived by a simple data transformation (similar to MASW), which maps a multimode dispersion curve. Figure 1c illustrates the processing steps of SSW, where *MinLen, MaxLen, MinOff*, and *MaxOff* will be defined below; *pos_x* is the in-line position of *v-f* determination; *trc_pos* is the trace position; *trc_off* is the source-receiver offset; *sel_trc* is the number of selected traces; and FFID is the field file identification number. One can observe that the main differences between MASW and SSW are windowing (which provides the possibility of stacking and enables higher horizontal resolution) and stacking (which enhances data quality by statistically increasing the signal-to-noise ratio).

Windowing parameters

Traces located at a predefined distance from a reference point are collected around a specific station from each possible record. After this, the *f-k* amplitude spectrum is calculated from each collection of traces. Optimization of the *f-k* spectrum requires two parameters (MinTraces, MaxTraces) to be determined.

1) MinTraces is the minimum number of traces to be collected from an appropriate record.
2) MaxTraces is the maximum number of traces to be collected from each record around a reference point.

These parameters each determine an interval of the profile; respectively, they can be called MinLength (MinLen) and MaxLength (MaxLen). This collection of data can be seen in Figure 2. Records 1 and 7 do not contain enough useful receivers for SSW, and record 8 is entirely out of

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the given interval. Increasing MaxLen decreases horizontal resolution, and decreasing MaxLen decreases data quality.

SSW expects parameter optimization at different frequencies because lower frequencies require longer spread and far offsets, although higher frequencies require near offsets. Two other offset parameters, MinOffset (MinOff) and MaxOffset (MaxOff), are required. Traces that fall into the interval determined by these two parameters are collected in SSW. Figure 3 shows the use of MinOff and MaxOff. Record 4 in Figure 2 originally contained nine useful receivers. Three of these nine receivers were rejected by windowing with MinOff (one geophone) and MaxOff (two geophones). Windowing with these parameters also resulted in a usable part of record 5 for SSW; however, record 6 could not be stacked because of the insufficient number of receivers.

Mixing of *v-f* planes

Mixing of different *v-f* planes is required, because wavelength of surface waves can vary over wide scales and each wavelength range requires different windowing parameters. Mixing guarantees high-quality data in the whole desired frequency range. The flowchart for mixing is shown in Figure 1d.

Fold in SSW

The fold in SSW is defined similarly to that in reflection stacking. Because the amplitude spectrum of a known number of traces is stacked at each place, the fold is the number of traces used to determine the dispersion curves at an individual point.

DEMONSTRATION OF THE PROPOSED SSW METHOD ON A FIELD DATA SET

The Pannonian GeoTraverse 4 (PGT-4) was an international, deep-seismic measurement designed to map the Mohorovicic Discontinuity (Hajnal et al., 1996). The measurement, using explosive sources of 50 kg dynamite at every third station, was carried out in the southeast part of Hungary. Geophone spacing of 100 m was used and the total length of the line was almost 100 km. Canadian PRS-type (Portable Recording System) low-frequency single geophones were used for recording. Each record was 60 s long, with 195 traces (19,400 m spread length). An example of the field records can be seen in Figure 4. Because the high quality, large-volume, low-frequency data set contains many separable modes, the PGT-4 data set is an appropriate choice for testing the algorithm.

Spatial aliasing

Because the measurement (PGT-4) was planned for deep seismic-reflection profiling, the horizontal sampling of the wavefield is relatively large for surface wave analysis. Therefore, we have to deal with the presence of spatial aliasing in this data set. Signals associated with $k_a > k_{Nyq}$ (k_a is the wavenumber of aliased signal and k_{Nyq} is the Nyquist wavenumber $1/2\Delta x$, where Δx is the receiver spacing) are aliased to $k_a - 2^* k_{Nyq}$ by the sampling theorem (Socco and Strobbia, 2004). In off-end configuration, the energy propagates in the positive direction and is associated with positive wavenumbers. Thus, the aliased signals can be recovered by $2^* k_{Nyq}$ unwrapping. Figure 5a and b shows the effect of this transformation on a shot

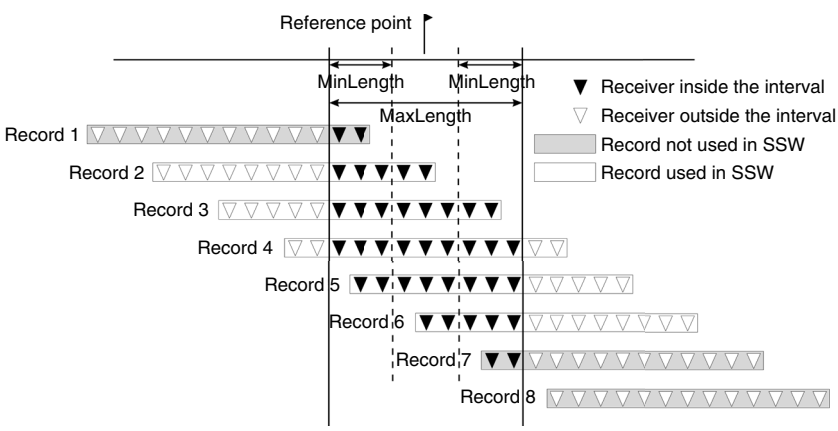

Figure 2. Simplified process of trace collection. MinLength and MaxLength each determine an interval of the profile. Records 1 and 7 do not contain enough useful receivers for SSW, although record 8 is out of the given interval. Increasing MaxLen decreases horizontal resolution.

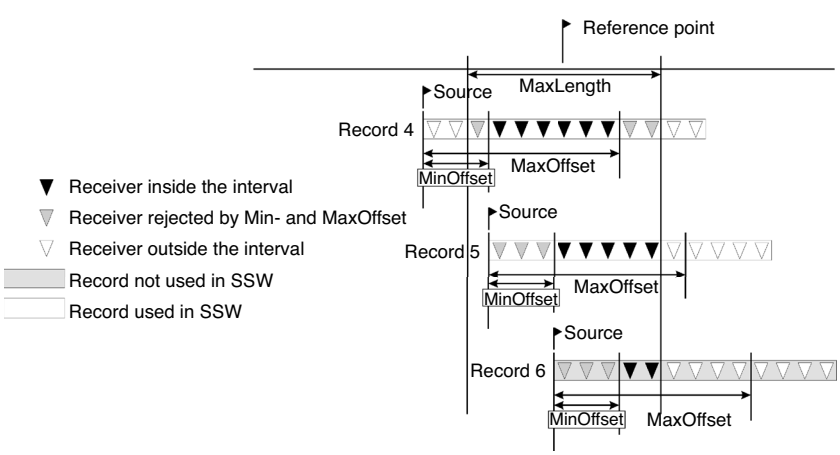

Figure 3. MinOffset and MaxOffset parameters. SSW expects parameter optimization at different frequencies because lower frequencies require longer spread and far offsets. Record 4 in Figure 2 contains nine useful receivers; after windowing with MinOff and MaxOff, three other receivers were rejected by MinOff (one geophone) and MaxOff (two geophones). These receivers are indicated in gray. Record 5 has enough usable traces for SSW, but record 6 cannot be used in stacking because of an insufficient number of receivers.

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record. Figure 5a shows the normal *f-k* field, which has positive and negative parts respectively, and Figure 5b contains only positive wavenumbers. Hence, the useful part of the *f-k* field is twice as large, but frequencies above 2.6 Hz are still spatially aliased.

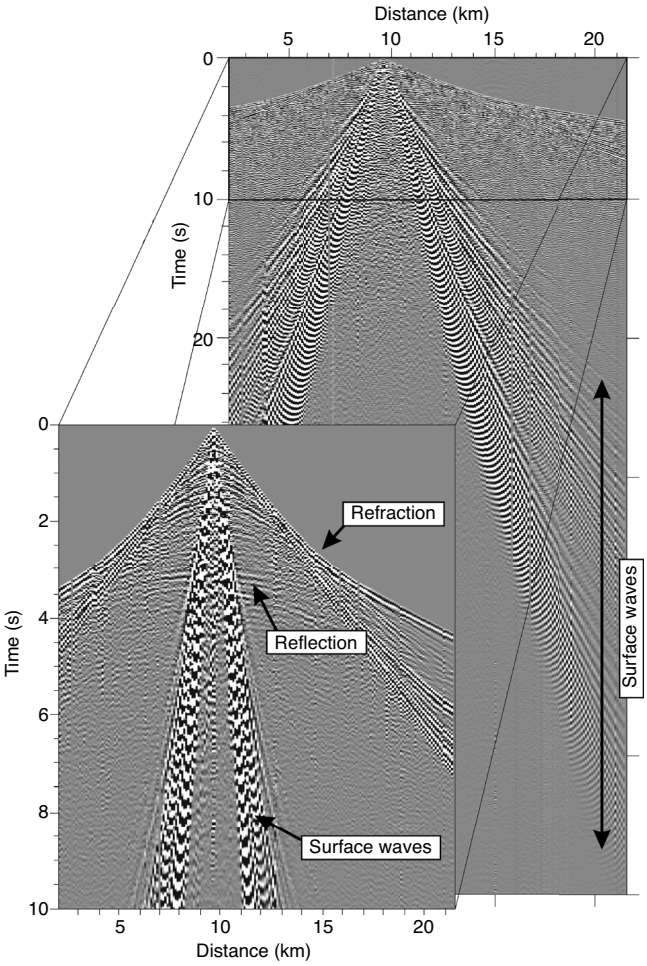

Figure 4. Field record with zoomed upper 10 s from PGT–4. Each seismogram has 195 traces, with 100-m geophone spacing and 60 s trace length. An explosive source of 50 kg dynamite was used at every third station. The complete surface wave is sampled on each record.

Yilmaz (1987) suggests three different approaches to recover from spatial aliasing: (1) Apply time shifts so that the steep (low velocity) events appear to have lower dips (higher velocities). (2) Apply a low pass filter to the data. (3) Apply a data-dependent interpolation scheme.

The second approach is not desirable because a significant part of the recorded frequency band is lost. Although the first approach is elegant, it increases the uncertainty, because a time shift transforms the high velocity signals to a lower wavenumber range. Hence, the third approach was chosen to solve the problem of spatial aliasing.

Data-dependent interpolation

An automatic mute was applied before interpolation to eliminate the unwanted high-velocity events at 5–10 Hz. These events are reflections from deep horizons and other coherent noises. Figure 5c shows the *f-k* amplitude spectrum after an automatic velocity mute was applied in the time domain. This is a simple and efficient method to remove other wave types.

To generate fill-in traces, data-dependent interpolation was applied to the data. This linear interpolation between adjacent traces was performed by scanning of possible velocities in the range of V_R –200–600 m/s with 25 m/s steps. In Figure 6a, part of a velocity-muted record is shown. Aliased events can be seen on high-frequency signals (depicted in this picture), which are recovered after data-dependent interpolation (Figure 6b). Three traces had been interpolated between each adjacent channel. The *f-k* spectrum and transformed *v-f* plots are shown in Figure 6d and g, and Figure 6e and h , respectively. The appropriate corners and boundaries are indicated in all domains. It can be seen that the aliased events were recovered in all domains of the interpolated record.

Effect of windowing

The array length affects the wavenumber resolution, and therefore the mode separation, as fewer receivers increase the uncertainties. The only (but very useful) advantage of windowing is the improved horizontal resolution. Figure 6c shows the spatially windowed part of the velocity-muted record in Figure 6b; Figure 6f and i contain the *f-k* and *v-f* domains of this reduced data set. The mode separation in Figure 6i is worse than in the case of non-windowed data (Figure 6h), but the horizontal resolution has improved (it will be verified later with another gather of data).

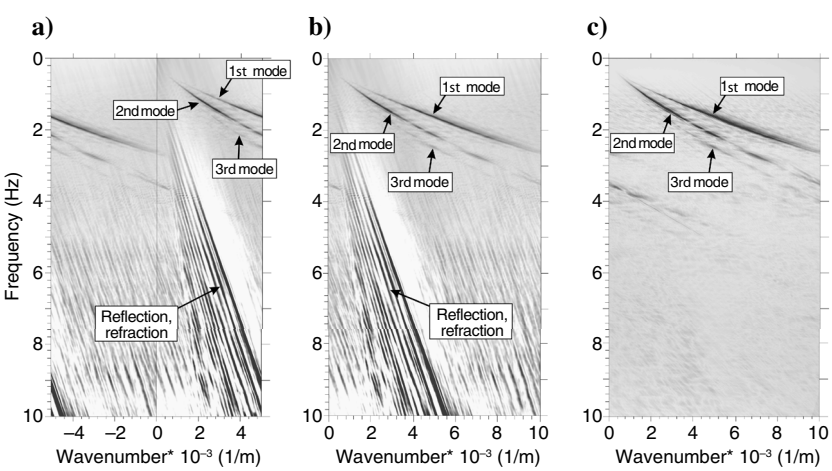

Figure 5. Signals associated with $k_a > k_{Nyq}$ (k_a is the wavenumber of aliased signal and k_{Nyq} is the Nyquist wavenumber $1/2\Delta x$, where Δx is the receiver spacing) are aliased to $k_a - 2^* k_{Nyq}$ by the sampling theorem. (a) Normal *f-k* field with positive and negative parts respectively; (b) the same *f-k* spectrum by $2^* k_{Nyq}$ unwrapping; (c) unwrapped *f-k* amplitude spectrum, where other than ground roll wave-types were removed by an automatic velocity mute.

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Stacking in the *f-k* domain

Figure 7 shows a collection of stacks. The *v-f* plots were enhanced by increasing the number of *f-k* amplitude spectra stacks at the same location (MinLen: 6000 m, MaxLen: 12,000 m). These high-quality *v-f* plots demonstrate the advantage of SSW compared with the MASW case (unstacked, top-left panel).

Optimization of windowing parameters

Figure 8 illustrates *v-f* plots with different MinOff (150 top row and 2000 m bottom row) and infinite MaxOff. Plots in the top row show significantly better results than the plots generated with MinOff = 2000 m. The lack of high frequency signals is obvious in the bottom row images.

Large uncertainty can be seen in Figure 8a (MinLen = 250 m, MaxLen = 500 m). In Figure 8b (MinLen = 500 m, MaxLen = 1000 m), the mode separation between the second and third modes is not clear. Although there is no significant difference between Figure 8c (1000 and 2000 m) and Figure 8d (2000 and 4000 m), 8c is a better result because it uses fewer traces and hence produces a better horizontal resolution.

Parameters (1000 and 2000 m) that produced the best result at the high frequency part are not suitable at low frequencies (Figure 9a) because their mode separation around 1 Hz is inadequate. Still, the calculation interval must be as small as possible because the local effects are otherwise obscured by surface waves from distant locations. The dispersion plot (Figure 9c), calculated using MinLen = 4000 m and MaxLen = 8000 m, provides a better result than the plots created from wider intervals in Figure 9d (8000 and 16,000 m) and narrower intervals in Figure 9b (2000 and 4000 m).

RESULTS

The most important consequence of the above examination is that higher and lower frequency surface waves require different processing parameters. The plot calculated with optimized parameters is shown in Figure 10. Between zero and 4 Hz, we chose a MinLen of 4000 m and a MaxLen of 8000 m. From 7 to 10 Hz, a MinLen of 1000 m and MaxLen of 2000 m were chosen. There is a mixed zone

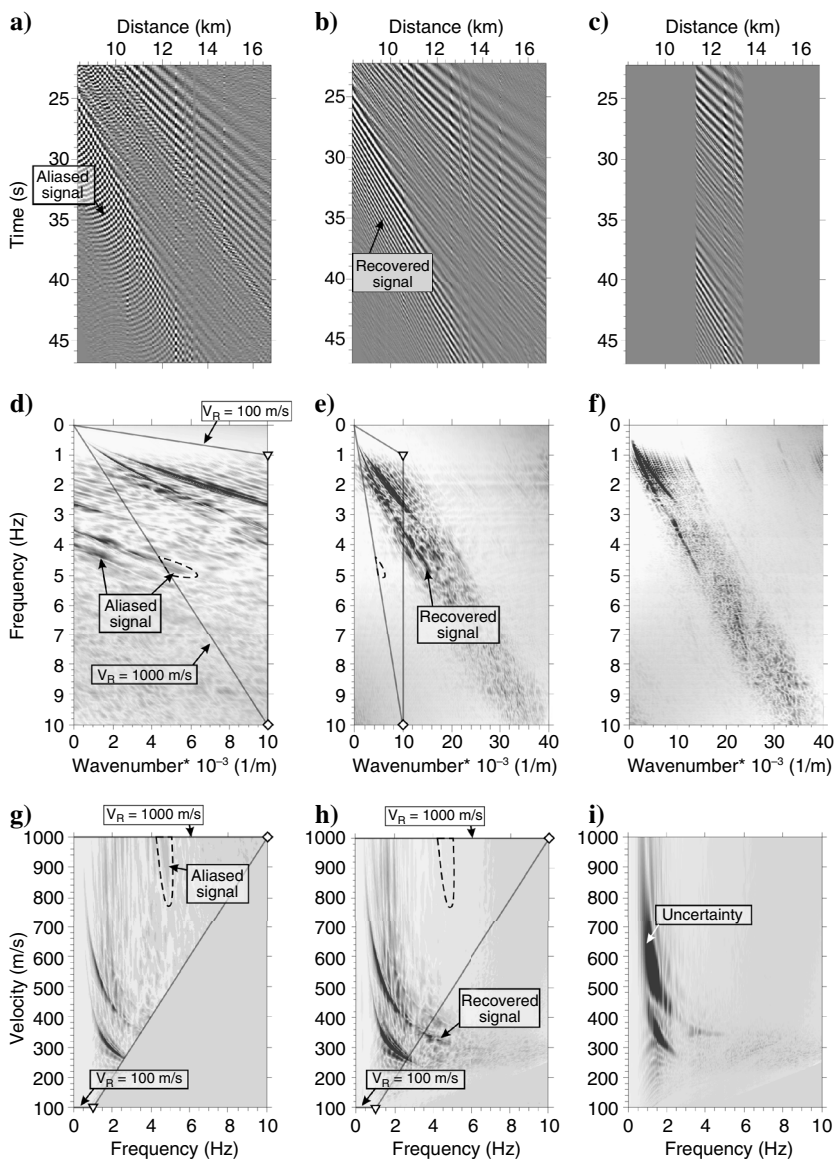

Figure 6. (a) Part of automatic muted field record; (b) record of (a), with data dependent interpolation; (c) space-domain windowed record of (b). (d-f) The *f-k* spectra of the records above them. (g-i) The *v-f* domain of the records above them. The appropriate corners and boundaries are indicated in all domains. Aliased events are recovered in all domains in the interpolated record. Uncertainty caused by time-domain windowing can be seen in (i).

between 4 and 7 Hz, with linearly changing weights. For better mode separation at low frequencies, a simple low-cut filter was applied to the image. Three modes can be easily separated, and the fourth mode can also be distinguished.

The horizontal resolution was checked by a specially ordered data set. Each v-f plot belongs to a station and therefore to an inline position. These plots can be located into a 3D coordinate system, where the vertical axis is the velocity (v); one of the two horizontal axes is the frequency (f); the other is the inline position (x). This 3D (f,v,x) cube can be sliced perpendicularly to the frequency axis (v, x plane). These slices are similar to the common offset gathers in reflection seismics; therefore they can be called *common-frequency gathers*. Two common frequency gathers, sliced at 2 (top) and 5 Hz (bottom), are shown in Figure 11. The section at 2 Hz has the first three well-

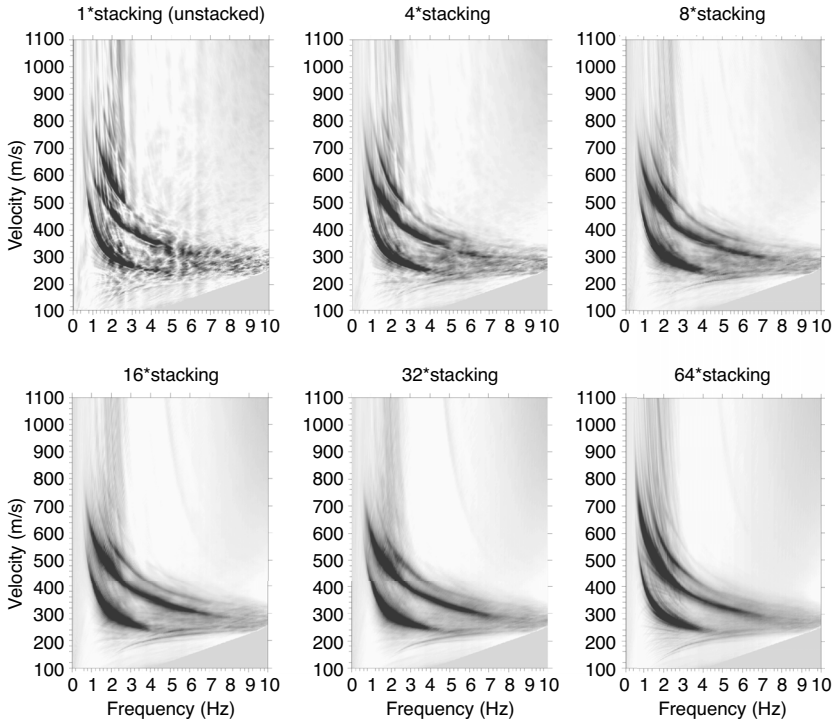

Figure 7. Effect of SSW with different number of records on v-f plots. The v-f plots are enhanced by increasing the number of f-k amplitude spectra stacks at the same place (MinLen: 6000 m and MaxLen: 12,000 m). These figures demonstrate the advantage of SSW compared with MASW (unstacked, in top-left panel) with the high-quality v-f plots.

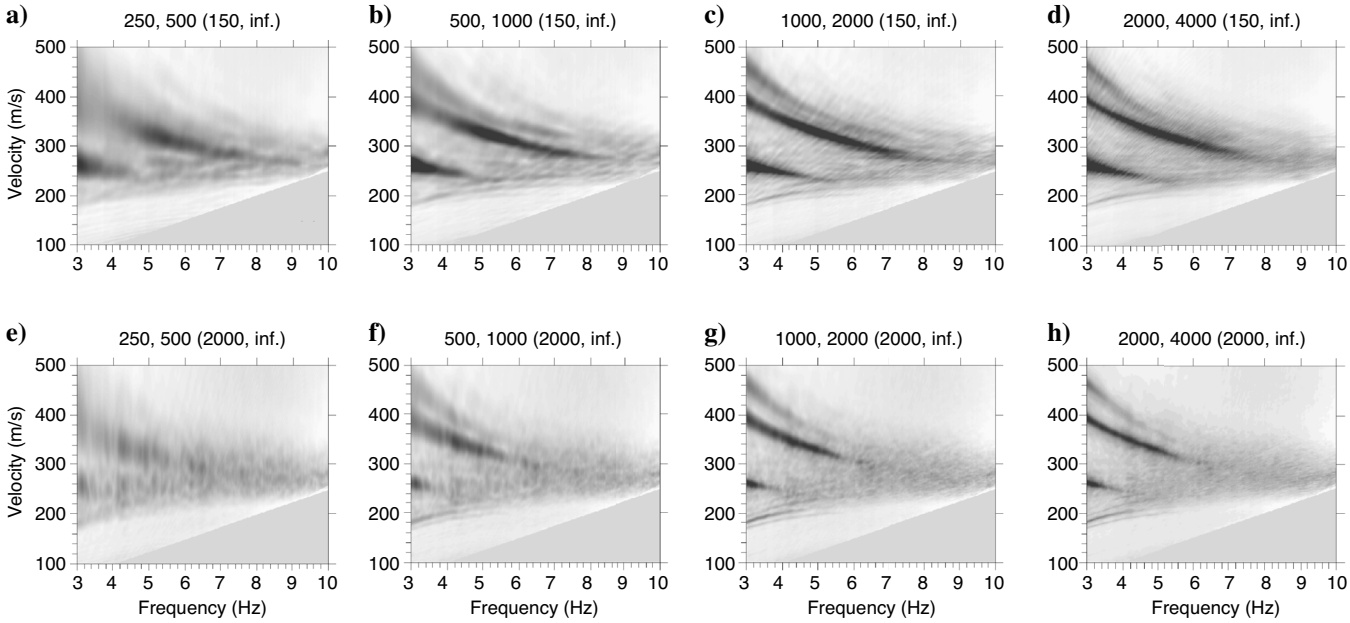

Figure 8. Effect of different parameters at the high frequency part of v-f domain. The top row illustrates results with MinOffset = 150 m, although the bottom row is calculated with MinOffset = 2000 m. MaxOffset is infinite in all cases. The first two numbers above the figures are the MinLength and MaxLength. A large uncertainty can be seen in (a), where MinLen = 250 m and MaxLen = 500 m, and the mode separation is not clear in (b) (500 and 1000 m) between the second and third modes. Although there is no significant difference between (c) (1000 and 2000 m) and (d) (2000 and 4000 m), (c) is a better result because it uses fewer traces and hence produces better horizontal resolution.

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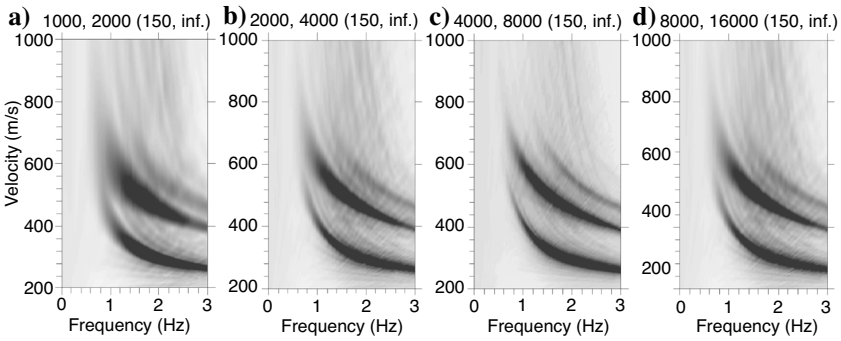

Figure 9. The effect of different MinLength and MaxLength parameters in SSW at low frequencies. (a) Parameters (1000 and 2000 m) that produced the best result at high frequencies are not suitable at low frequencies because the mode separation around 1 Hz is inadequate. (b) Dispersion plot created from narrower intervals (2000 and 4000 m). (c) This dispersion plot (calculated using MinLen = 4000 m and MaxLen = 8000 m) provides a better result than plots (b) and (d). (d) Dispersion plot created from wider intervals (8000 and 16,000 m).

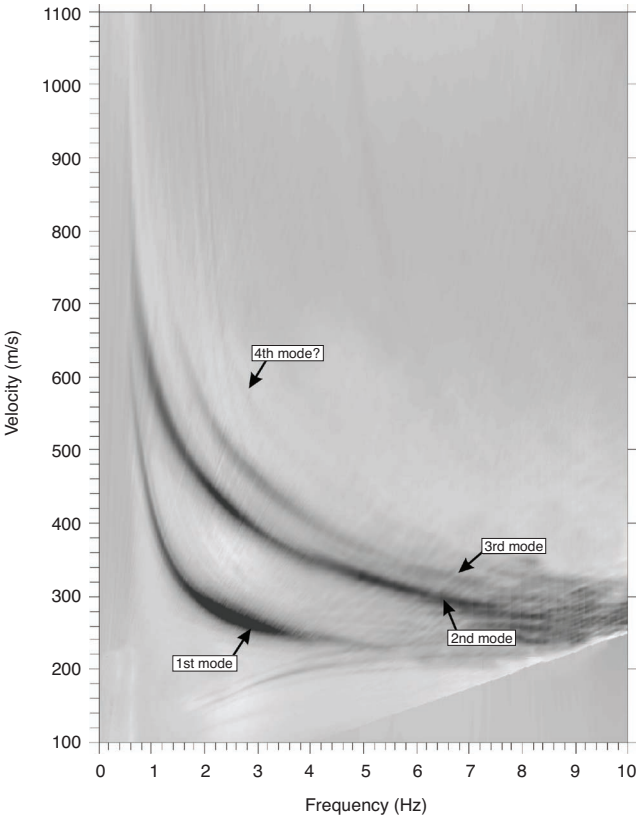

Figure 10. v-f plot determined by mixed parameters. Between zero and 4 Hz, MinLen = 4000 m and MaxLen = 8000 m, whereas from 7 to 10 Hz, the parameters are MinLen = 1000 m and MaxLen = 2000 m. Between 4 and 7 Hz, there is a mixed zone, with linearly changing weights. Three modes can easily be separated, and the fourth mode can also be distinguished.

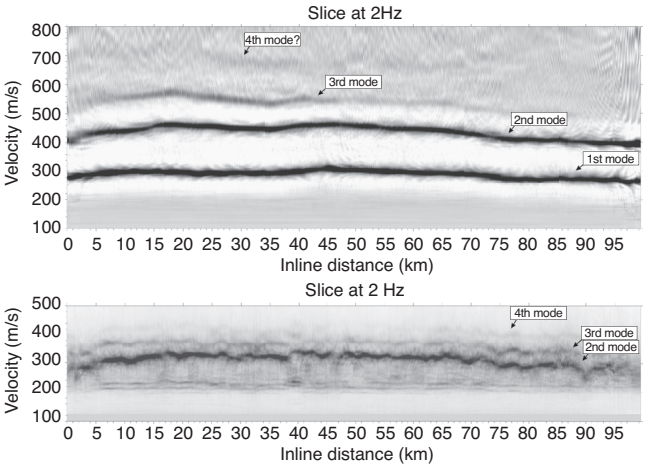

Figure 11. Horizontal resolution was checked by a specially ordered data set. Each v-f plot belongs to a station and therefore to an inline position. These plots can be located into a 3D coordinate system, where the vertical axis is the velocity (v); one of the two horizontal axes is the frequency (f), the other is the in-line position (x). This 3D (f,v,x) cube can be sliced perpendicularly to the frequency axis (v, x plane). The section at 2 Hz has the first three well-separable modes. The section at 5 Hz does not contain the first mode (see Figure 10), but the fourth mode appears.

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separable modes. The section at 5 Hz does not contain the first mode, but the fourth mode appears. Horizontal variation with a wavelength of approximately 3000–4000 m is shown mainly in the 5 Hz section by the second mode.

CONCLUSIONS

Stacking of surface waves is a new method proposed for obtaining multimode dispersion curves, based on stacking of records in the *f-k* domain. This new approach was developed because traditional techniques have two important limits: sensibility to data quality and horizontal resolution. The horizontal resolution of the MASW method was increased by space-domain windowing, which in turn decreased data quality. Stacking improved data quality; hence the number of traces transformed into the *f-k* domain could be decreased, although the possibility of mode separation was preserved. High-quality dispersion curves were obtained using parameters optimized by analysis.

When the geophone spacing in the deep-seismic data was too large (Δx = 100 m) for surface waves, preprocessing (mute and space domain interpolation) salvaged this data set. This suggests that the real power of the proposed method could be best demonstrated by applying it to a data set with appropriate station interval and array length. Stacking of surface waves can be used for 3D measurements as well and is also applicable for any near-surface P-S_V (Rayleigh wave) and S_H (Love wave) seismic measurements. The advantages and applicability of this new approach are illustrated by means of theoretical investigations as well as a case study example.

ACKNOWLEDGMENTS

The author would like to thank the Eötvös Loránd Geophysical Institute for providing the seismic data. Special thanks also to his colleagues — especially László Hermann and Tamás Bodoky — for their helpful ideas and support.

REFERENCES

Hajnal, Z., B. Reilkoff, K. Posgay, E. Hegedus, E. Takacs, I. Asudeh, St. Mueller, J. Ansorge, and R. DeIaco, 1996, Crustal-scale extension in the central Pannonian basin: Tectonophysics, **264**, 191–204.
McMechan, G. A., and M. J. Yedlin, 1981, Analysis of dispersive waves by wave field transformation: Geophysics, **46**, 869–874.
Nazarian, S., and K. H. Stokoe II, 1984, In situ shear wave velocity from spectral analysis of surface waves: Proceedings of the 8th World Conference on Earthquake Engineering, vol. 3, 31–39.
Nolet, G., and G. F. Panza, 1976, Array analysis of seismic surface waves: Limits and possibilities: Pure and Applied Geophysics, **114**, 776–790.
Park, C. B., R. D. Miller, and J. Xia, 1999, Multichannel analysis of surface waves: Geophysics, **64**, 800–808.
Socco, L. V., and C. Strobbia, 2004, Surface-wave method for near-surface characterization: A tutorial: Near Surface Geophysics, **2**, 165–186.
Stokoe II, K. H., S. G. Wright, J. A. Bay, and J. M. Roesset, 1994, Characterization of geotechnical sites by SASW method, in R. D. Woods, ed., Geophysical characterization of sites (ISSMFE TC#10): Oxford & IBH Publishing Co., 15–25.
Yilmaz, O., 1987, Seismic data processing: SEG

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